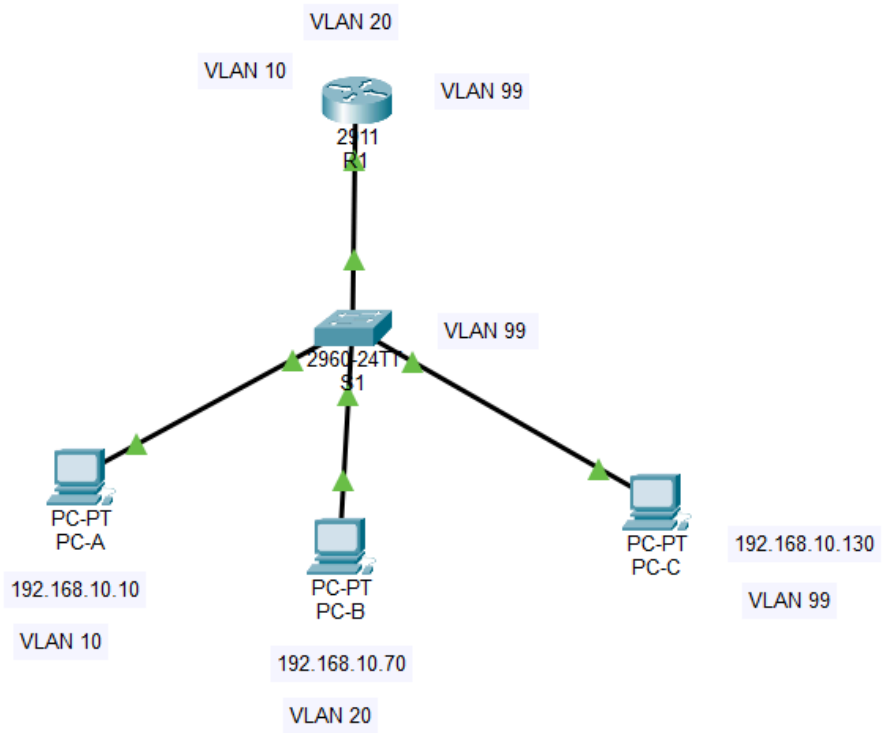
LAB 2 REPORT : Secure & Scalable VLAN Design (Router-on-a-Stick) ***Last Updated: 16 Dec 2025***

▪TOPOLOGY DIAGRAM (บอส / ปวริศช์ ประมวล / 673380278-9)

None

```
                    [ R1 – Router ]
                    G0/0 (802.1Q Trunk)
                              |
                              |
                     [ G0/1 – Switch ]
                    ┌─────────┴─────────┐
                    |         S1        |
                    |         |         |
            [Fa0/1]       [Fa0/2]       [Fa0/3]
                |         |         |
                |         |         |
              [PC-A] [PC-B]         [PC-C]
              VLAN10  VLAN20        VLAN99
```



▪ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
- **Subnet Mask (/26):** 255.255.255.192
- **Subnets Used:**
  - VLAN 10 → 192.168.10.0/26
  - VLAN 20 → 192.168.10.64/26
  - VLAN 99 → 192.168.10.128/26

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

▪LAB INSTRUCTION STEPS

▫PART 1: SWITCH CONFIGURATION (บอส / ปวริศช์ ประมวล / 673380278-9)

Step 1: Basic Switch Setup

```
None
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
```

Step 2: Create VLANs

```
None
S1(config)# vlan 10
S1(config-vlan)# name USERS
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name SERVERS
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name MANAGEMENT
S1(config-vlan)# exit

S1(config)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config-vlan)# exit
```

```
S1(config)#vlan 10
S1(config-vlan)#name USERS
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#exit
S1(config)#vlan 99
S1(config-vlan)#name MANAGEMENT
S1(config-vlan)#exit
S1(config)#vlan 999
S1(config-vlan)#name BLACKHOLE
S1(config-vlan)#exit
```

Step 3: Secure Unused Ports (Security First)

```
None
S1(config)# interface range fa0/4-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

```
S1(config)#interface range fa0/4-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 999
S1(config-if-range)#shutdown
```

```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S1(config-if-range)#exit
```

Step 4: Assign Access Ports

```
None
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/3
```

```
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#
S1(config-if)#no shutdown
S1(config-if)#exit

S1(config)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#no shutdown
S1(config-if)#exit

S1(config)#interface fa0/3
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 99
S1(config-if)#no shutdown
S1(config-if)#exit
```

Step 5: Configure Trunk to Router

```
None
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#interface g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,20,99
S1(config-if)#no shutdown
S1(config-if)#exit
```

Step 6: Configure Management SVI

```
None
S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# ip default-gateway 192.168.10.129
```

```
S1(config)# end
S1# copy running-config startup-config
```

```
S1(config)#interface vlan 99
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
ip address 192.168.10.131 255.255.255.192
S1(config-if)#
S1(config-if)#no shutdown
S1(config-if)#exit

S1(config)#ip default-gateway 192.168.10.129
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

▫PART 2: ROUTER CONFIGURATION (บอส / ปวริศช์ ประมวล / 673380278-9)

Step 7: Basic Router Setup

```
None
Router> enable
Router# configure terminal
Router(config)# hostname R1
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Step 8: Enable Physical Interface

```
None
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
```

Step 9: Configure Router-on-a-Stick Subinterfaces

None

```
R1(config)# interface g0/0.10
R1(config-if)# encapsulation dot1Q 10
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.20
R1(config-if)# encapsulation dot1Q 20
R1(config-if)# ip address 192.168.10.65 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.99
R1(config-if)# encapsulation dot1Q 99
R1(config-if)# ip address 192.168.10.129 255.255.255.192
R1(config-if)# exit
```

```
R1(config)#interface g0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit

R1(config)#interface g0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.10.65 255.255.255.192

R1(config)#interface g0/0.99
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.10.129 255.255.255.192
R1(config-subif)#exit
```

None

```
R1(config)# end
R1# copy running-config startup-config
```

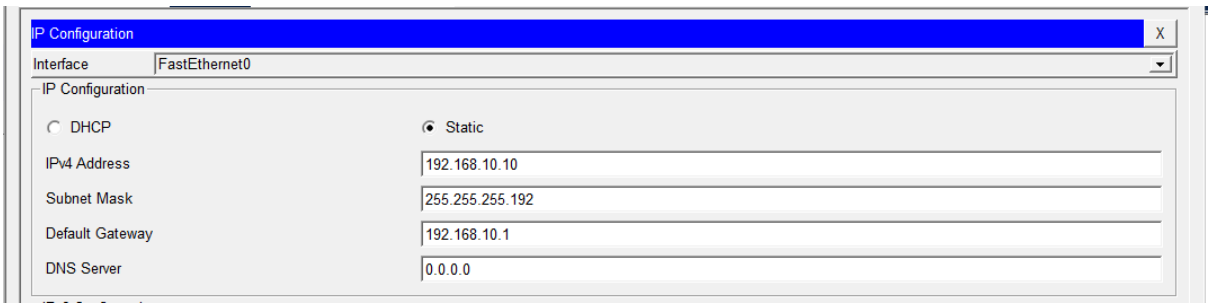
```
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

▫PART 3: END DEVICE CONFIGURATION (673380515-1 ศรัณย์ พาพรชัย (อ๋อง))

Configure each PC manually.

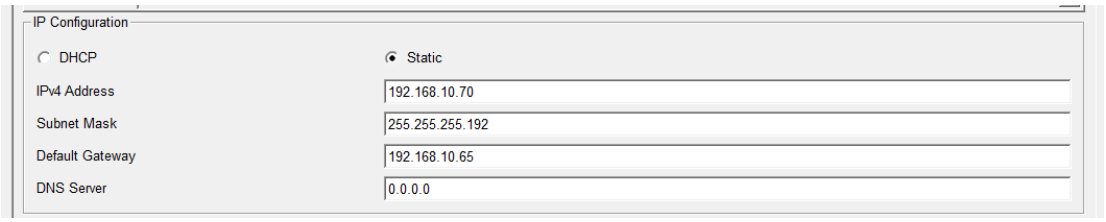
PC-A (VLAN 10)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1



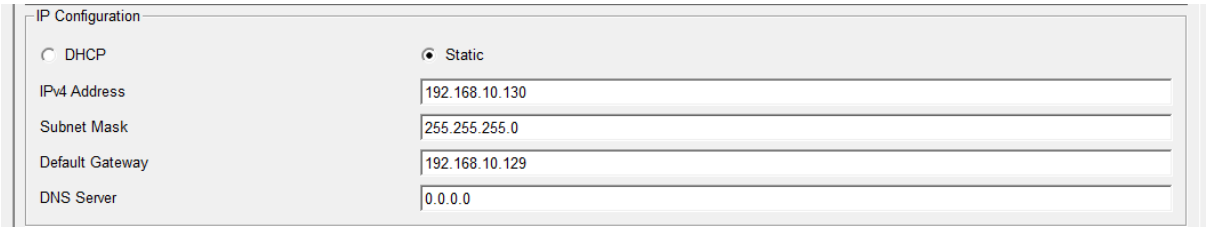
PC-B (VLAN 20)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65



PC-C (VLAN 99)

- IP: 192.168.10.130
- Mask: 255.255.255.192
- Gateway: 192.168.10.129



▫PART 4: CONNECTIVITY VERIFICATION

Step 10: Verification Commands

From S1

```
None
show vlan brief
show interfaces trunk
```

```
S1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
90   VLAN0090                         active
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                               Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                               Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                               Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                               Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                               Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

From R1

```
None
show ip interface brief
```

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.10.65   YES manual up                    up
GigabitEthernet0/0.99  192.168.10.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

From PCs

```
None
ping 192.168.10.1
ping 192.168.10.70
ping 192.168.10.131
```

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Expected:

- PC-A ↔ PC-B ✅
- PC-A ↔ S1 Management IP ✅

▫PART 5: TROUBLESHOOTING CHALLENGE

Scenario A: Wrong VLAN Assignment (673380515-1 ศรัณย์ พาพรชัย (อ๋อง))

- Move PC-B to the wrong VLAN

```
S1(config-if)#switchport access vlan 1
S1(config-if)#exit
S1(config)#exit
S1#
```

Change the switchport that connected to PC-B to vlan 1 (default) instead of vlan 20

- Test connectivity

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC-A and PC-C cannot ping to PC-B

- Identify failure point

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active
55   VLAN0055                         active
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

The show vlan brief command has shown port Fa0/2 (PC-B) active in the default vlan (1)

- Restore correct VLAN

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
55   VLAN0055                         active
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

Can be fixed with command switchport access vlan 20 (correct vlan for PC-B)

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk

```
S1(config)#interface g0/1
S1(config-if)#switchport trunk allowed vlan remove 99
```

- Test access to S1

```
Vlan99                 192.168.10.131  YES manual up                    up
```

```
C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Explain why management fails

  The ping fails because VLAN 99, which carries S1's management traffic, is not allowed on the trunk, so management frames cannot traverse the link..

- Restore configuration

```
S1(config)#interface g0/1
S1(config-if)#switchport trunk allowed vlan add 99
```

Scenario C: Wrong Gateway (บอส / ปวริศช์ ประมวล / 673380278-9)

- Change PC-A gateway

from

Default Gateway	192.168.10.1

to

Default Gateway	192.168.10.9

- Observe ping failure

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Use `ipconfig` and `ping`

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::240:BFF:FE70:76CB
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.9

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

```
C:\>ping 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Fix gateway

  from

  Default Gateway 192.168.10.9

  to

  Default Gateway 192.168.10.1

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

▪LAB DELIVERABLES

Worksheet 1: Configuration Check (673380515-1 ศรัณย์ พาพรชัย (อ๋อง))

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	Found all vlan in the switch (10, 20, 99)	Pass ✅
Trunk active	show int trunk	VLANs allowed	Vlans allowed on trunk for Port Gig0/1	Pass ✅
Router subifs	show ip int br	up/up	Gig0/0/0.10, Gig0/0/0.20, Gig0/0/0.99 All up status	Pass ✅
Inter-VLAN ping	ping	Success	PC can ping each other successful	Pass ✅

VLANs exist :

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                               Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                               Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                               Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                               Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                               Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Trunk active :

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

Router subifs :

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    up
GigabitEthernet0/0/0.20192.168.10.65   YES manual up                    up
GigabitEthernet0/0/0.99192.168.10.129  YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

PC-A → PC-B

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time=1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-B → PC-C

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time=1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

PC-C → PC-A

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Worksheet 2: Design Understanding (673380293-3 ศิระพัทธ์ วงศ์วิวัฒน์เสรี)

Answer briefly:

1. Why is VLAN 99 used for management?
   VLAN 99 is commonly used as a **management VLAN** to keep device management traffic separate from user data traffic.
2. Why is VLAN 999 assigned to unused ports?
   VLAN 999 is often used as a **parking VLAN** for unused switch ports
3. What does 802.1Q tagging do?
   802.1Q tagging allows **multiple VLANs to travel over a single trunk link**
4. Why does each VLAN require a different subnet?
   Each VLAN is a **separate broadcast domain**, so it must use a different IP subnet

---

▪VERIFICATION COMMANDS CHECKLIST

Router

- [x] ~~show run~~

```
R1#show run
Building configuration...

Current configuration : 979 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
 --More--
```

- [x] ~~show ip route~~

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.10.64/26 is directly connected, GigabitEthernet0/0.20
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0.20
C       192.168.10.128/26 is directly connected, GigabitEthernet0/0.99
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0.99
```

- [x] ~~show arp~~

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.10.10           11  0001.4204.A822  ARPA   GigabitEthernet0/0.10
Internet  192.168.10.70           10  0002.1682.9E9E  ARPA   GigabitEthernet0/0.20
Internet  192.168.10.130          10  0010.11E0.A030  ARPA   GigabitEthernet0/0.99
```

Switch

- [x] ~~show vlan brief~~

```
S1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
90   VLAN0090                         active
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                               Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                               Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                               Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                               Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                               Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

- [x] ~~show interfaces trunk~~

```
S1>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

- [x] ~~show mac address-table~~

```
S1>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.2f1b.dd01    DYNAMIC     Gig0/1
  10    0060.2f1b.dd01    DYNAMIC     Gig0/1
  20    0060.2f1b.dd01    DYNAMIC     Gig0/1
  99    0060.2f1b.dd01    DYNAMIC     Gig0/1
```

PCs

- [x] ~~ipconfig~~

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::210:11FF:FEE0:A030
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.130
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.129

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

- [x] ~~arp -a~~

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.70         0060.2f1b.dd01        dynamic
  192.168.10.129        0060.2f1b.dd01        dynamic
  192.168.10.131        0001.c798.5401        dynamic
```

- [x] ~~ping~~

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

▪ ASSESSMENT QUESTIONS (673380272-1 ธนภูมิ จันทรา (โชกุน))

1. Which device performs inter-VLAN routing?
    - **The R1 Router**, VLAN 10, 20 and 99 separate the network into distinct broadcast domains. For traffic to travel from a device in VLAN 10 to a device in VLAN 20.

2. Why can PCs in different VLANs communicate?
    - **Because of the inter-VLAN routing performed by R1**, When PC-A sends a packet destined for PC-B, PC-A recognizes that the destination is outside its local subnet so it sends the traffic to its Default Gateway.

3. What breaks if the trunk is removed?
    - **Inter-VLAN communication will stop immediately**. Without the trunk, the switch has no way to forward frames tagged with VLAN 10, 20, or 99 to R1 for routing.

4. Why is VLAN 1 avoided?
    - **For security reasons**, because VLAN 1 is so universal, it is a primary target during attacks. By using custom VLAN IDs like 10, 20, 99, it is harder for simple attack scripts to target sensitive network traffic and devices.

5. What security risk exists if unused ports remain active?
    - **Unauthorized access and physical security breach**, An active, unused port provides a live connection point. If an attacker gains physical access to it, they can simply plug a device into the active port and immediately gain access to the default VLAN's network resources.